

Daniel Afkhami

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SKILLS

- **Programming:** Python, Java, TypeScript/JavaScript, HTML/CSS, SQL, LaTeX, Bash
- **Frameworks & Libraries:** React, Tailwind CSS, Vite, FastAPI, NumPy/SciPy, PyTorch, PySide6
- **Databases:** PostgreSQL, Supabase
- **Cloud & Tools:** Git, Docker, Azure, Vercel, Railway, Addepar

EXPERIENCE

Software Engineer Intern

May 2026–August 2026

Stone Temple Partners, LLC

New York, NY

Full-stack engineer on a wealth-management platform for RIAs serving UHNW households.

- Implemented CRM functionality in use by 8 advisors overseeing \$750M+ AUM, designing **RESTful** endpoints using **FastAPI** against a **React/TypeScript** frontend and a **Supabase/PostgreSQL** schema.
- Shipped a portfolio analysis and tax-aware rebalancing engine that scores a prospective client's portfolio against a Monte Carlo optimizer and generates executable transition plans with capital-gains netting, loss harvesting, and wash-sale constraints.
- Delivered a Windows desktop application (**PySide6**) that records advisor calls, transcribes them via **Azure** AI Speech, and files LLM-generated summaries to five CRM record types, cutting transcription costs ~95% by retiring a third-party service.
- Built automated ETL pipelines (Supabase Edge Functions on **pg_cron**) that reconcile position-level data via the **Addepar** API, giving advisors direct access to household AUM, cash balances, and per-fund performance for pooled vehicles.
- Replaced password authentication with Microsoft **OAuth 2.0** SSO and built an in-app mail client via the **Microsoft Graph** API with automated email filing, eliminating stored credentials and provisioning every Microsoft integration from one sign-in.

EDUCATION

Rutgers University - Honors College

Piscataway, NJ

Bachelor of Science in Computer Science and Mathematics | GPA: 3.94/4.00 (Dean's List)

Expected May 2029

- **Coursework:** Systems Programming, Data Structures, Computer Architecture, Mathematical Theory of Probability, Real Analysis, Elementary Differential Equations
- **Activities:** Colorstack, Rutgers University Ski Team, Rutgers Symphonic Winds, Rutgers Cars

Metuchen High School

Metuchen, NJ

GPA: 4.51/4.00 (Summa Cum Laude) | SAT: 1540 (800 Math, 740 Reading & Writing)

Graduated June 2025

PROJECTS

Denosing Diffusion Probabilistic Model (DDPM) Synthesizing 2D Ising System Configurations

April 2026–August 2026

- Implemented a **NumPy** Monte Carlo engine to generate a dataset of Ising model lattice states at critical temperatures.
- Designed a DDPM using **PyTorch** and a **U-Net** architecture in Python to predict spin configurations at critical temperatures.
- Validated performance by statistically comparing generated energy/magnetization distributions against theoretical baselines.

Physics-Informed Neural Network (PINN) Modeling Field-Induced Motion of Ferrofluid

August 2024–September 2025

- Derived governing force equations for the movement of ferrofluid droplets through viscous water under the influence of field gradients, and utilized them to numerically compute traversal times of ferrofluid droplets toward a magnetic field source.
- Designed and implemented a PINN using **TensorFlow** and **Scikit-Learn** libraries in Python to predict traversal times.
- Achieved accurate extrapolation from limited data through enforcing residuals of the governing differential equation.